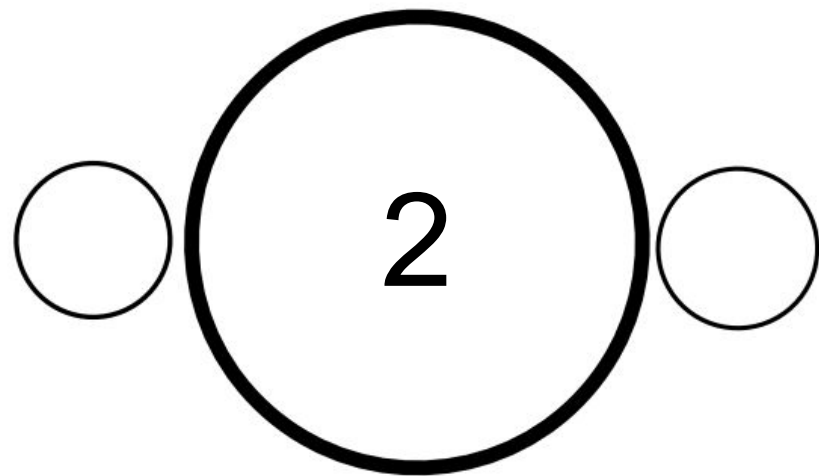
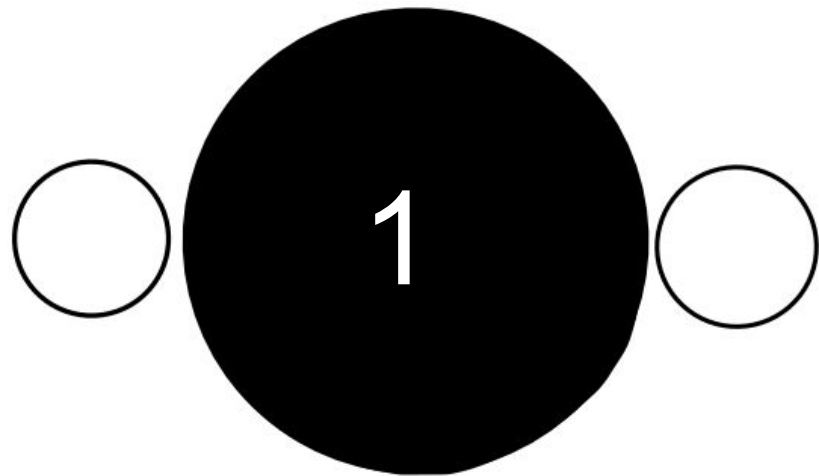


Microscale Reactions

*Neutralising citric acid and
precipitating lead iodide.*





Precipitating lead iodide

Method

1. Add potassium iodide to the left small circle.
2. Add lead nitrate to the right small circle.
3. Add 10 drops of water into the large central circle.
4. Push the crystals into the edges of the water.
5. Observe the dissolution of the crystals and the formation of lead iodide.

Big scale! - we not allowed :(



Neutralising Citric Acid

Method

1. Add anhydrous sodium carbonate to the left small circle.
2. Add citric acid to the right small circle.
3. Add 10 drops of water into the large central circle.
4. 1 drop of indicator solution to the central drop of water.
5. Push the crystals into the edges of the drop of water.



Big scale!

What's happening in the Precipitation of PbI_2 ?

Potassium is more reactive than lead.

So it pushes lead out of its compound.

When it is heated, the yellow precipitate will **disappear** and the solution will become **colourless***

*However, we cannot do this (**Very dangerous**)
insert crying emoji

THE EXPLANATION



When they “**swap places**” PbI_2 is formed.

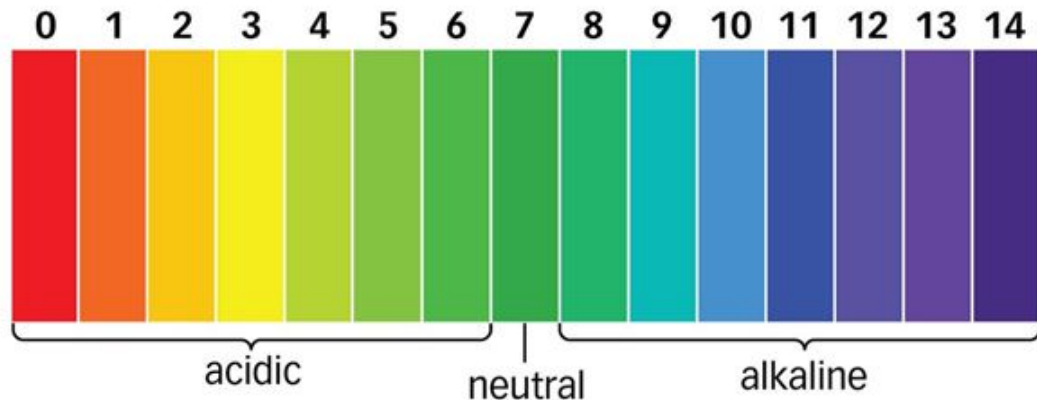
Lead iodide is **more soluble** in **warm** water than **cold**!

When the solution **cools down**, crystals of lead iodide begin to form, causing the ‘**golden rain**’ effect.

What's happening in the Neutralisation of Citric Acid?

You have put an **acidic** substance in the **right** small circle and an **alkaline** substance in the **left** small circle.

Since you have added **water** in the middle, there will be a **fusion** of the acid and the alkali.



When there is a **fusion** between an acid and an alkali, **neutralisation** takes place.

So you will get some **very different colours** depending on **how much** acid or alkali is there or if a fusion has taken place.